

Oceanic zones

Covering nearly three-quarters of Earth's surface, and reaching down to 9 miles (11 km) at their deepest point, the oceans make up the biggest biome by volume. All life here lives submerged in salty marine waters, but conditions vary enormously from the coastlines down to the ocean's bottom.

Sunlit zone

(0–650 ft/0–200 m)

Bright sunlight provides energy for ocean food chains that start with algae.

Twilight zone

(650–3,280 ft/200–1,000 m)

Sunlight cannot penetrate far into the ocean. As depth increases, conditions are too dark for algae, but animals thrive.

Midnight zone

(3,280–13,000 ft/
1,000–4,000 m)

Animals find different ways of surviving in the dark ocean depths. Many use bioluminescence: they have light-producing organs to help them hunt for food or avoid danger.

Abyssal zone

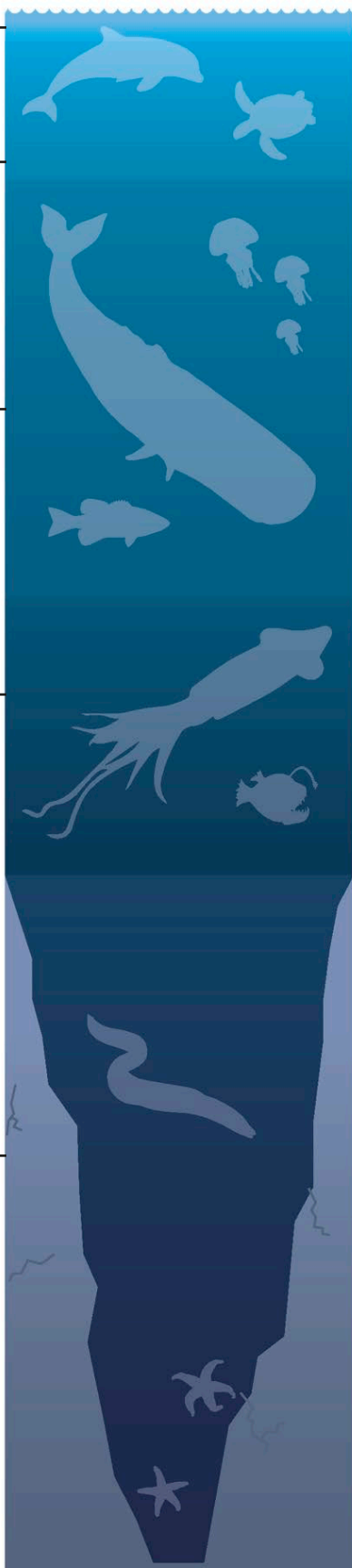
(13,000–19,650 ft/
4,000–6,000 m)

Near the ocean floor, water pressure is strong enough to crush a car and temperatures are near freezing. Most food chains here are supported by particles of dead matter raining down from above.

Hadal zone

(19,650–36,000 ft/
6,000–11,000 m)

The ocean floor plunges down into trenches that form the deepest parts of the ocean. But even here there is life—with a few kinds of fishes diving down to 26,000 ft (8,000 m) and invertebrates voyaging deeper still.

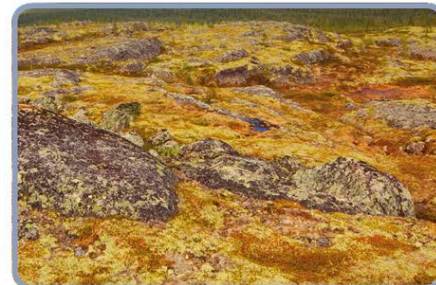


Biomes

Places exposed to similar sets of conditions—such as temperature or rainfall—have similar-looking habitats, even when they are as far apart as North America and Asia. These habitat groups are called biomes. Over continents and islands, they include tundra, deserts, grasslands, forests—and freshwater lakes and rivers.

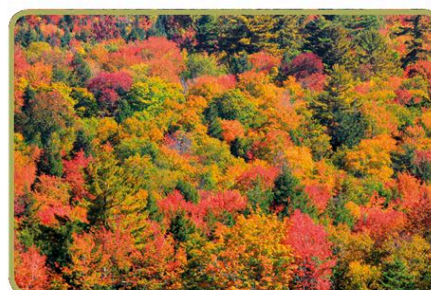
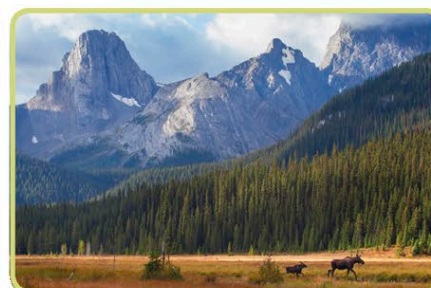
Tundra

Where land is close to the poles, conditions are so cold that the ground is permafrost—meaning it is frozen throughout the year. Here, trees are sparse or cannot grow at all, and the thin vegetation is made up of grasses, lichens, and small shrubs.



Taiga

The largest land biome is a broad belt of coniferous forest that encircles the world below the Arctic tundra. Conifers, pines, and related trees have needlelike leaves that help them survive low temperatures. They are evergreen—so they retain their tough foliage even in the coldest winters.



Temperate forest

The Earth's temperate zones are between the cold polar regions and the tropics around the equator. Many of the forests that grow in these seasonal regions are deciduous: they produce their leaves during the warm summers, but lose them in the cold winters.

Temperate grassland

Where the climate is too dry to support forests but too wet for desert, the land is covered with grassland—a habitat that supports a wide range of grazing animals.

Temperate grasslands experience seasonal changes in temperature, but stay green throughout the year.



Tropical dry and coniferous forest

Some tropical regions have pronounced dry seasons that can last for months. Here, many kinds of trees drop their leaves in times of drought. Others have adaptations that help them to stay evergreen. In places, the forests are dominated by conifers with drought-resistant leaves.

